SYSTEMS AND METHODS FOR INTRALUMINAL COMPLIANCE AND DISTANCE PROFILING USING TIME-OF-FLIGHT RANGING, ULTRASONIC SENSING, AND CONTROLLED EXPANSION MEASUREMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. Provisional Patent Application No. __63/983,946______, filed ___02/16/2026 03:21:42 PM Z ET_____, the contents of which are incorporated herein by reference in their entirety.

In certain embodiments, the systems and methods described herein may operate independently or as part of a multi-node physiologic measurement platform comprising biometric sensing devices, anatomical geometry acquisition systems, longitudinal intraluminal monitoring architectures, calibration subsystems, compatibility modeling engines, encrypted profile generation frameworks, machine learning synthesis modules, or user-specific personalization infrastructures.

The present disclosure is directed to intraluminal compliance and distance profiling architectures operable as standalone devices while maintaining interoperability with coordinated system deployments.

In various implementations, the disclosed device corresponds to a sensing node configured to cooperate with additional sensing nodes within a unified physiologic data ecosystem, enabling correlation with anatomical acquisition systems, longitudinal monitoring devices, wearable biometric nodes, or distributed computational analysis platforms configured to generate composite user profiles.

The disclosure supports implementations in which the device operates independently, alongside complementary sensing hardware, or as part of an integrated multi-device physiologic measurement system operating under shared computational analysis.

FIELD OF THE INVENTION

The present disclosure relates generally to biometric measurement systems and intraluminal anatomical profiling technologies. More specifically, the disclosure relates to systems and methods for generating a user-specific intraluminal compliance and distance profile using time-of-flight ranging, ultrasonic distance sensing, controlled expansion measurement, pressure response characterization, and computational profile synthesis techniques.

In various embodiments, the disclosure pertains to portable, handheld, body-insertable, or session-based devices configured to obtain quantitative measurements within a body lumen under standardized positioning and controlled measurement conditions.

The systems described herein may incorporate ranging modules configured to determine distance between a distal sensing region and an internal anatomical boundary in combination with compliance sensing mechanisms configured to characterize deformation response, expansion tolerance, pressure distribution, or dynamic tissue interaction behavior.

In certain embodiments, the device may be configured for use within vaginal, anal, or other intraluminal anatomical environments while maintaining applicability to broader luminal measurement contexts.

Distance acquisition may be performed using optical time-of-flight sensors, LiDAR-based ranging systems, ultrasonic transducers configured for acoustic time-of-flight measurement, or combinations thereof, while compliance characterization may be achieved through pressure sensing elements, inflatable expansion regions, strain measurement structures, monitored pump output, or hybrid sensing approaches.

Measured data may be processed through computational algorithms configured to generate structured intraluminal profiles representing spatial distance characteristics, compliance behavior, response curves, and derived physiological indices suitable for longitudinal analysis, calibration, or integration into broader biometric analytics systems.